Expanding Single Brackets

KS3 Algebra · Year 7/8 · Multiplying out brackets

Name: ________________________ Class: __________ Date: __________

HOW TO EXPAND

Multiply everything inside the bracket by the term outside. Work through each term one at a time. Watch out for negative signs — a negative outside the bracket changes the sign of every term inside.

WORKED EXAMPLES

Basic

$2(x + 3) = 2 \times x + 2 \times 3 = 2x + 6$

Coefficient of 1

$1(x + 3) = 1 \times x + 1 \times 3 = x + 3$

The hidden 1 means the bracket stays the same when expanded.

Harder coefficient

$3(2x - 1) = 3 \times 2x + 3 \times (-1) = 6x - 3$

Negative outside

$-2(x + 4) = -2 \times x + (-2) \times 4 = -2x - 8$

The sign belongs to the term.

$-3x - 5x$ means $-3x + (-5x) = -8x$

$3x - (-5x) = 3x + 5x = 8x$

Remember: neg × pos = neg, neg × neg = pos.

Q1–Q15 — BASIC

1) $2(x + 3)$

2) $5(x + 1)$

3) $3(x - 2)$

4) $4(2x + 1)$

5) $4(x + 3)$

6) $6(x - 3)$

7) $7(x + 3)$

Q1–Q15 CONTINUED

9) $9(x - 1)$

10) $3(x + 5)$

11) $10(x + 2)$

12) $4(x - 2)$

13) $6(x + 1)$

14) $5(x - 3)$

15) $8(x + 3)$

8) $2(x + 5)$

Expanding Single Brackets — Practice

Expand each expression, then simplify where possible

Q16–Q30 — INTERMEDIATE

16) $7(3x + 2)$

17) $-2(x + 5)$

18) $-3(4x - 1)$

19) $5(2x - 3)$

20) $7(2x - 4)$

21) $-3(2x - 3)$

22) $5(3x + 4)$

23) $-4(x + 3)$

24) $4(3x - 2)$

25) $-5(x - 6)$

26) $6(3x + 2)$

27) $-7(x + 3)$

28) $4(2x + 5)$

29) $-9(x - 3)$

30) $8(3x - 2)$

Q31–Q45 — ADVANCED

31) $-4(3x + 2)$

32) $6(2x - 5)$

33) $-5(x - 7)$

34) $2(3x^2 + 4x)$

35) $-5(2x - 3)$

36) $7(2x - 3)$

37) $-4(2x^2 - 3x)$

38) $6(3x + 4)$

39) $-3(2x^2 + 3x)$

40) $5(4x - 7)$

41) $2(2x^2 + 3x - 5)$

42) $-5(3x - 4)$

43) $6(2x^2 - 3x)$

44) $-4(x^2 + 2x - 3)$

45) $5(2x^2 + 5x - 3)$

SPOT THE MISTAKE

For each one, write the **correct answer** and a short **reason** explaining the mistake.

Amy writes: **$3(2x + 3) = 6x + 3$**

Correct answer:

Reason:

Ben writes: **$-2(x + 4) = -2x + 8$**

Correct answer:

Reason:

Cara writes: **$-3(2x - 5) = -6x - 15$**

Correct answer:

Reason: